Rakamin X id/x partners

CREDIT RISK ANALYSIS

Final Task of ID/X Partner Data Scientist
Project Based Internship
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Hallo, my name is

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I'm an Informatics student in my final semester with a strong passion for data. I'm particularly interested in data analysis and machine learning, while currently exploring the broader world of data science and data engineering. I'm eager to contribute to real-world projects where I can apply my analytical thinking and grow through hands-on experience.

OUR AGENDA

1. Introduction: Project Background & Objectives
2. Methodology: Data & Analysis Workflow
3. Key Findings: Exploratory Data Analysis (EDA)
4. Modeling: Our Approach & Algorithms
5. Model Evaluation: Performance Comparison
6. Conclusion & Recommendations

THE CHALLENGE: CREDIT RISK IN MULTIFINANCE

In the multifinance industry, accurate credit risk assessment is fundamental to profitability.

- **The Problem: Failure to accurately identify high-risk borrowers (those likely to default).**
- **The Impact: Increased Non-Performing Loans (NPLs), leading to significant financial losses and instability.**
- **The Need: A data-driven, objective, and highly accurate system to evaluate credit risk.**

PROJECT OBJECTIVES

Our primary goal for this project is:
To develop and evaluate a robust machine learning model that accurately predicts the probability of a borrower defaulting (Bad Credit).

Key Targets:
- Provide a data-driven decision support tool for loan approval.
- Identify the key features and patterns that signal high credit risk.
- Reduce potential financial losses by improving the early detection of high-risk loans.

OUR DATA SCIENCE WORKFLOW

Data Understanding
- Load 2007-2014 historical loan data.

Data Preparation
- Data Cleansing (Handling NaNs, removing unneeded columns).
- Target Engineering (Creating the binary good_bad class).

Feature Engineering
- Converting date features into numeric durations (in months).
- Cleaning and encoding term and emp_length.

Handling Imbalance
- Applying SMOTE to the training dataset.

Modeling
- Training 3 algorithms (Logistic Regression, Random Forest, XGBoost).

Evaluation & Recommendation
- Selecting the best model based on business-critical metrics.

DATA AT A GLANCE

- Source: Historical Loan Data (2007 - 2014)
- Initial Size: 466,285 loan records, 75 features.
- Final Data (After Prep): 239,010 clean records for modeling.

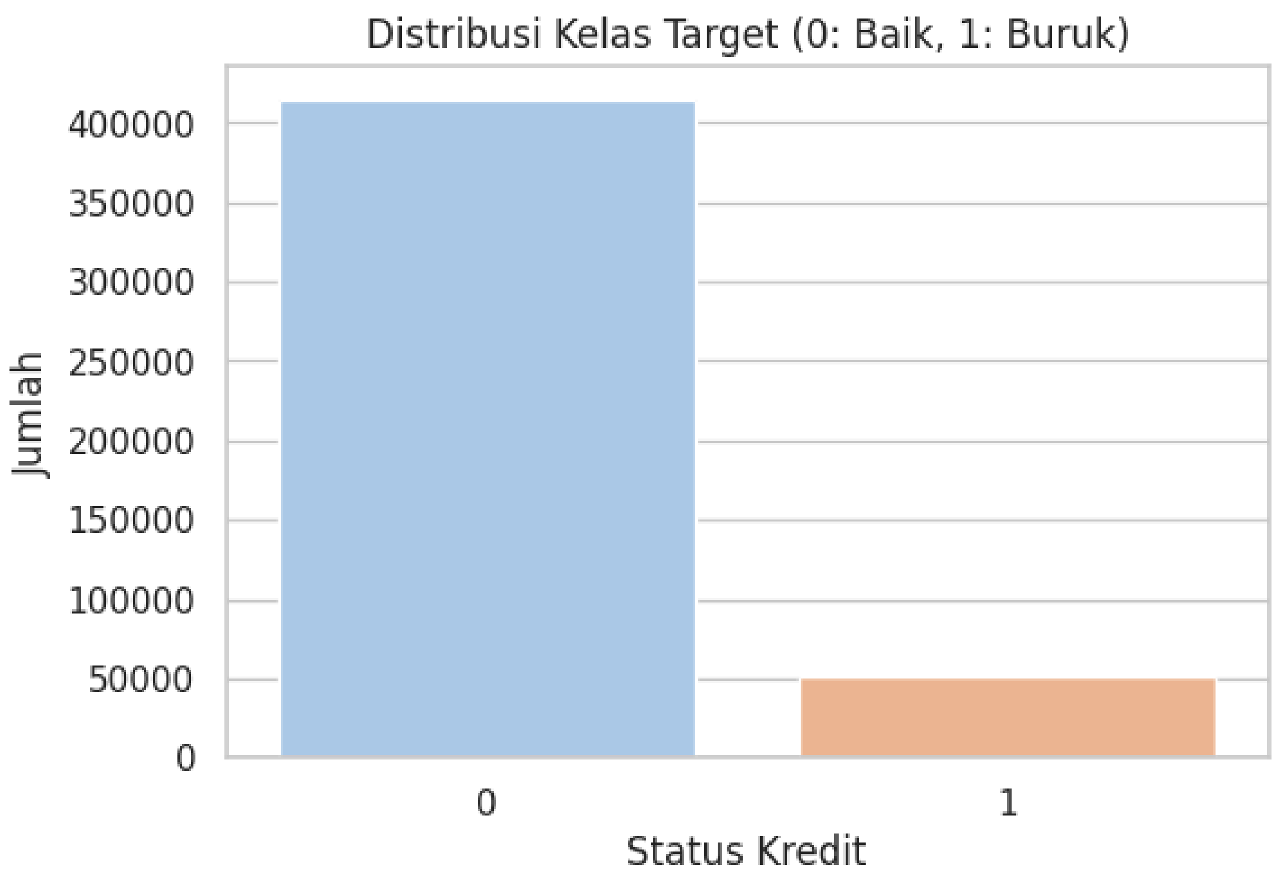
Target Feature Engineering:

The original loan_status (9 categories) was ambiguous for modeling. We engineered a clear, binary target:

- Bad Credit (1): 'Charged Off', 'Default', 'Late (31-120 days)', etc.
- Good Credit (0): 'Fully Paid', 'Current', 'In Grace Period', etc.

Key Challenge: Severe Class Imbalance

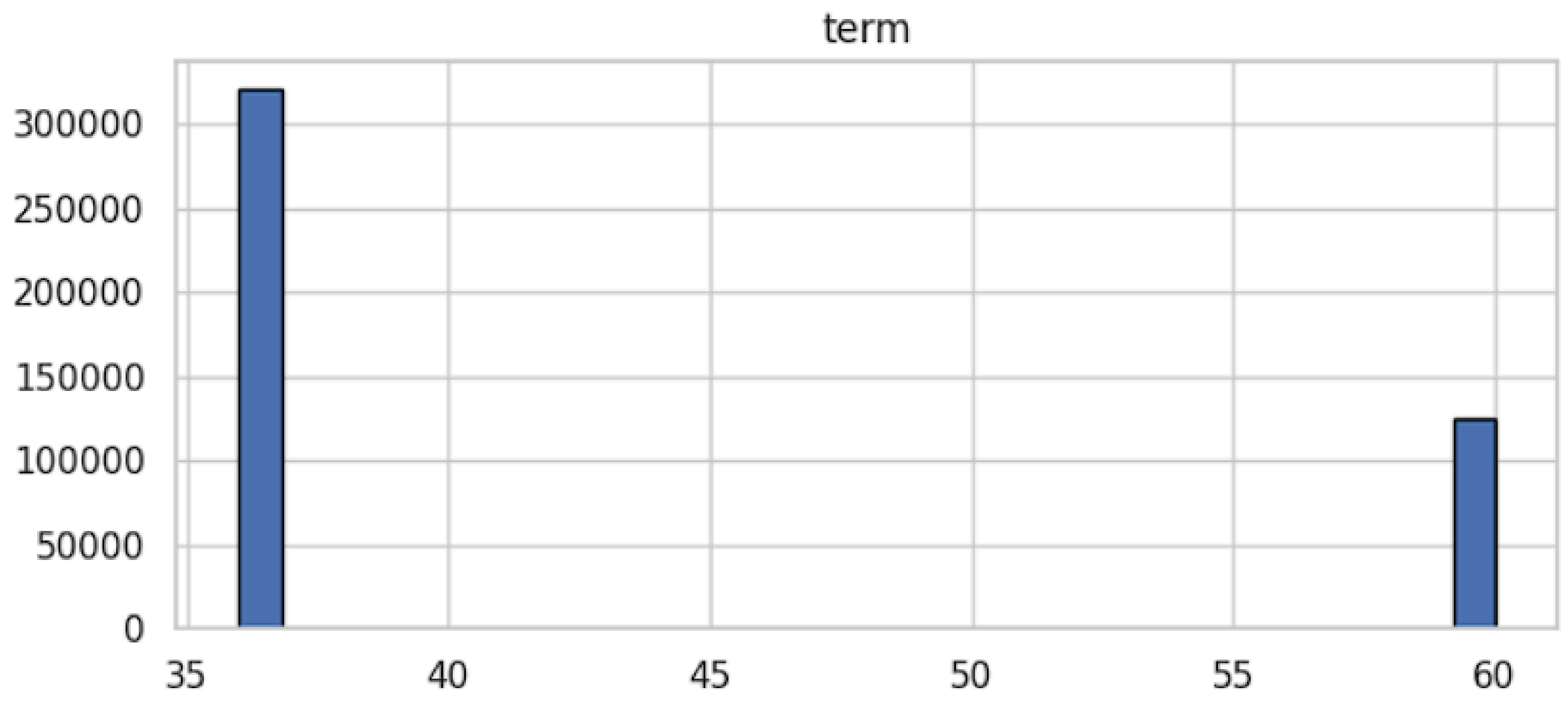
- Our clean dataset was highly imbalanced.
- This must be solved before modeling to prevent a biased result.



WHAT THE DATA TOLD US

Exploratory Data Analysis (EDA) revealed critical insights:

- Significant Growth: A sharp, exponential increase in loan issuance, peaking in 2014.
- Popular Term: The majority of borrowers (approx. 70%) consistently choose the 36-month loan term.
- Credit History Matters (ANOVA): Our ANOVA test confirmed a statistically significant difference (p-value < 0.05) in the average 'Credit Age' between the 'Good' and 'Bad' credit groups.



PREPARING DATA FOR MODELING

To make our data "speak" to the models, we performed two key steps:

Date Conversion:

- Date columns like earliest_cr_line, issue_d, and last_pymnt_d are useless as text.
- We converted them into numeric features representing "Months Since..." (e.g., mths_since_issue_d). This measures duration, which is meaningful.

Feature Scaling:

- All 6 of our final numeric features were standardized using StandardScaler.
- This ensures all features (e.g., term vs. mths_since_earliest_cr_line) are treated with equal importance by the models.

SOLVING THE CLASS IMBALANCE PROBLEM

- **Problem:** If we train on 89% "Good" data, the model will be lazy. It will get 89% accuracy by just predicting "Good" every time, failing to find the "Bad" loans we care about.
- **Solution: SMOTE** (Synthetic Minority Over-sampling Technique).
- **How it Works**: We apply SMOTE only to the training data. It intelligently creates new, "synthetic" examples of the minority class ('Bad Credit').
- **Result:** A perfectly balanced 50/50 training set, forcing the model to learn the patterns of both good and bad borrowers.

THE COMPETING MODELS

We trained and compared three powerful and distinct classification algorithms:

1. **Logistic Regression**
   - A fast, interpretable, and industry-standard baseline model.
2. **Random Forest**
   - A robust Ensemble (Bagging) model. It builds hundreds of decision trees and uses their collective vote, which prevents overfitting.
3. **XGBoost (eXtreme Gradient Boosting)**
   - A state-of-the-art Ensemble (Boosting) model. It builds trees sequentially, with each new tree correcting the errors of the last.

PERFORMANCE SCORECARD

Models were evaluated on the unseen Test Set.

Model	Accuracy	Precision (Bad)	Recall (Bad)	F1-Score (Bad)	ROC-AUC
Logistic Regression	850	94	**74**	83	850
Random Forest	**954**	98	**93**	**95**	**954**
XGBoost	942	**99**	90	94	942

Key Observations:

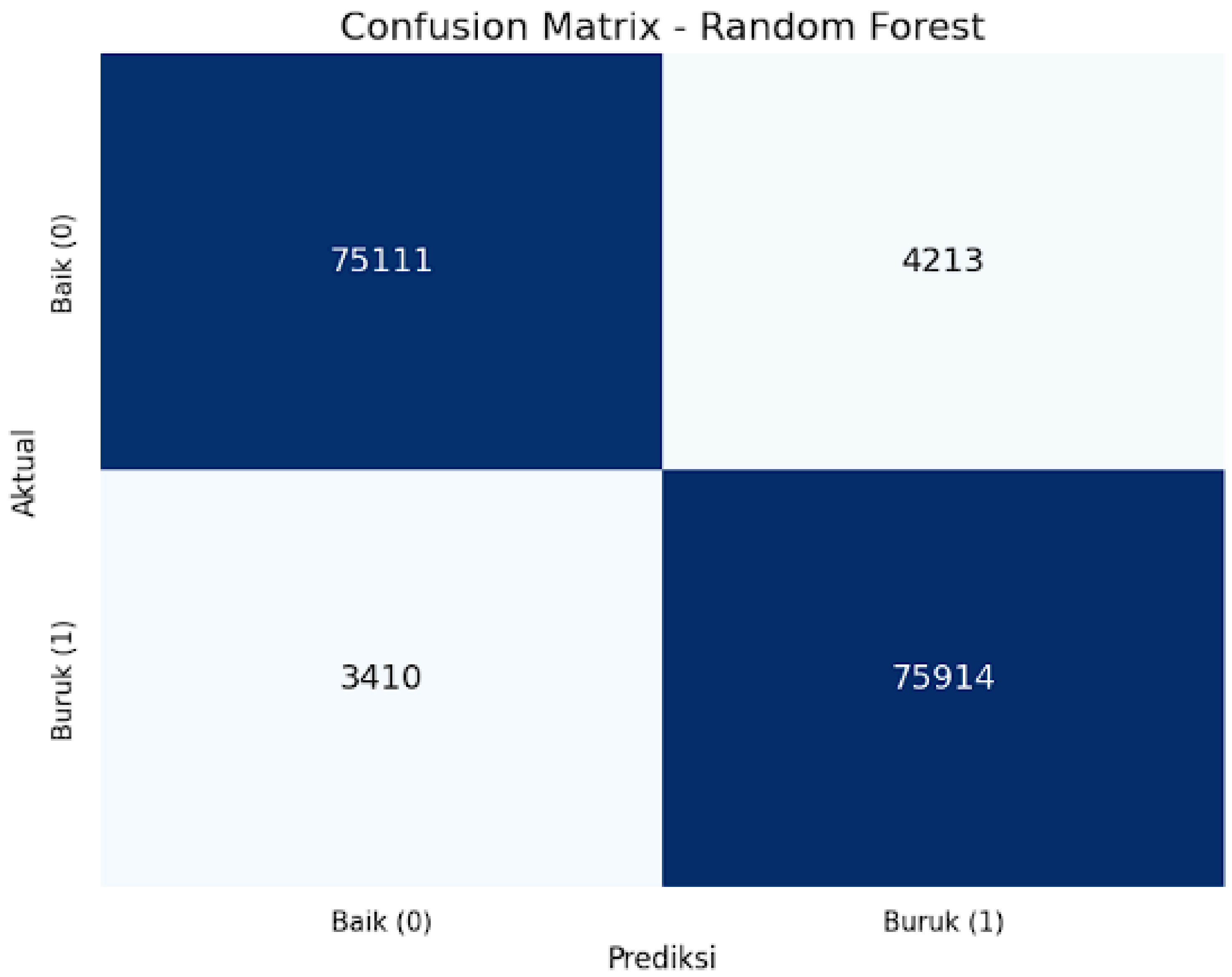
- **Logistic Regression** has dangerously low Recall (74%)—it misses 26% of all bad loans.
- **XGBoost** has near-perfect Precision (99%)—when it flags a loan as 'Bad', it's almost certainly correct.
- **Random Forest** has the highest Recall (93%) and the best-balanced F1-Score (0.95), plus the highest ROC-AUC (0.954).

THE BUSINESS COST OF ERRORS

In credit risk, not all errors are created equal.

- **Type 1 Error (False Positive):**
  - **What:** Model predicts 'Bad', but the customer is actually 'Good'.
  - **Business Cost:** Lost opportunity (a good customer is denied).
- **Type 2 Error (False Negative):**
  - **What:** Model predicts 'Good', but the customer is actually 'Bad'.
  - **Business Cost:** Financial Loss! (The loan defaults).

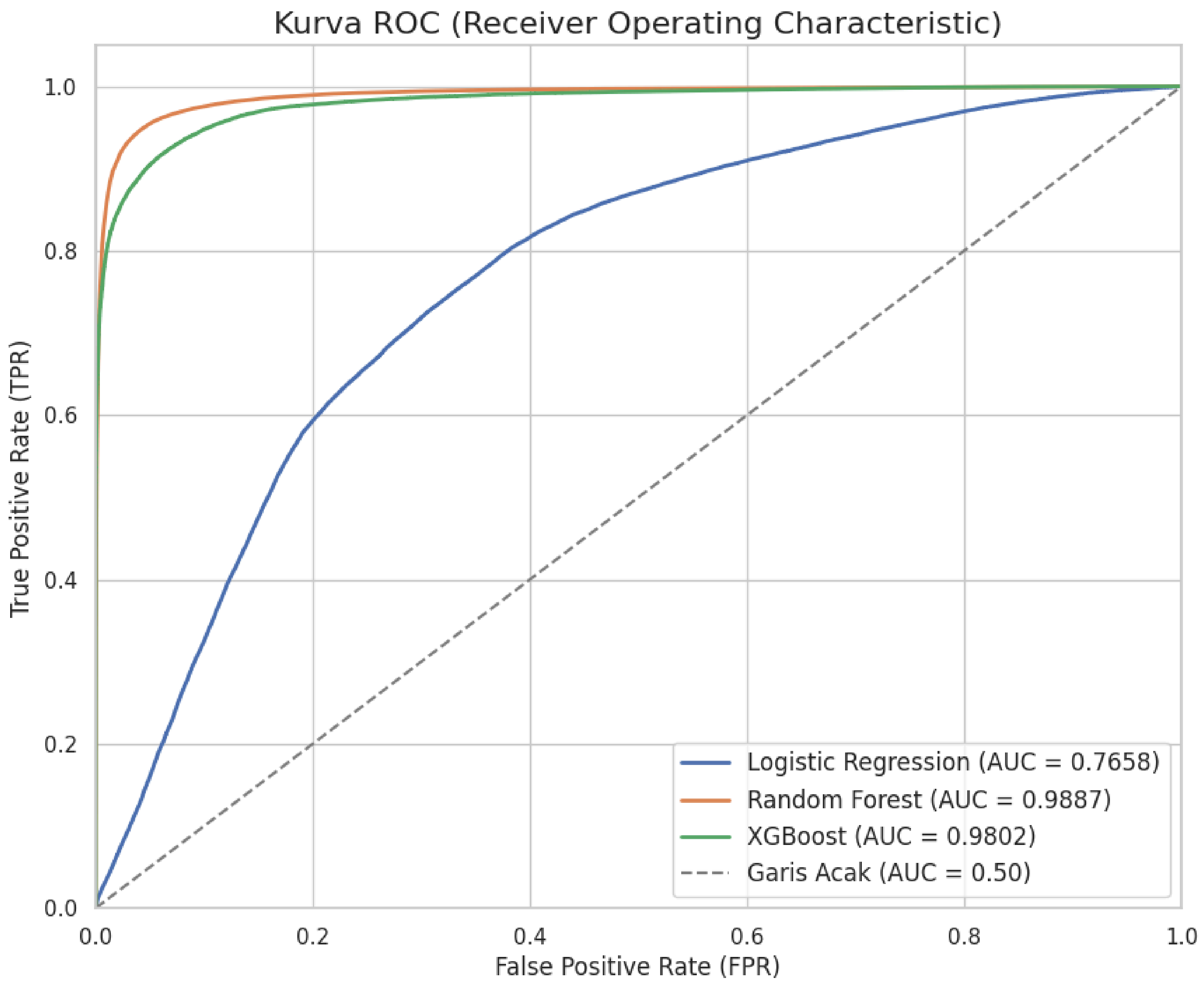
Our Priority: We must minimize False Negatives, which means maximizing Recall.



MODEL DISCRIMINATORY POWER (ROC-AUC)

The ROC curve shows how well a model can distinguish between 'Good' and 'Bad' credit.

- The closer the curve is to the top-left corner (AUC = 1.0), the better.
- **Random Forest (AUC = 0.954)** demonstrates the best overall ability to separate the two classes, outperforming the other models.



FINAL RECOMMENDATION

Based on the analysis and the critical business need to identify bad loans:

1. **Logistic Regression (Not Recommended):** Performance is insufficient. A **74% Recall** is too low for this high-stakes problem, as it misses one in four bad loans.
2. **XGBoost (Strong Candidate):** An excellent choice if the only goal is to be extremely certain about 'Bad' predictions (99% Precision). However, it misses more bad loans (90% Recall) than our top choice.
3. **Random Forest (Recommended Model):**
   - This model provides the **best balance** of all metrics.
   - With the **highest Recall (93%) and highest F1-Score (0.95)**, it is the most effective model at achieving the primary goal: **finding the most bad loans** while remaining highly accurate.
   - Its highest AUC (0.954) confirms it is the best classifier overall.

FUTURE ENHANCEMENTS

To further improve our risk assessment capabilities, we recommend:

- **Hyperparameter Tuning:** Conduct a deeper, more granular tuning (e.g., GridSearch, RandomizedSearch) of the Random Forest model to extract maximum performance.
- **Feature Importance Analysis:** Analyze the Random Forest model to understand which features (e.g., credit age, term, last payment date) are the most powerful predictors.
- **Cross-Validation:** Re-evaluate the final model using K-Fold Cross-Validation for a more robust and reliable performance estimate.
- **Cost-Sensitive Learning:** Explore advanced techniques that explicitly tell the model that a False Negative is "more expensive" than a False Positive.

THANK YOU

GitHub: Credit Risk Analysis